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GOVERNMENT OF INDIA
OFFICE OF THE DIRECTOR GENERAL OF CIVIL AVIATION

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INSTALLATION, CALIBRATION AND MAINTENANCE REQUIREMENTS OF AVIATION METEOROLOGICAL EQUIPMENT AT PUBLIC AERODROMES

1. Introduction

1.1 Aviation Meteorological services play a vital role in ensuring the safety and efficiency of civil aviation operations. The regulatory requirements for meteorological service provider are outlined in Civil Aviation Requirements (CAR) Section 9 Series M Part I.

1.2 India Meteorological Department (IMD) is responsible for providing aeronautical meteorological services for safety and regularity of air navigation in India

1.3 Weather factors have significant influence on the operation and performance of aircraft. The impact of a relatively small change in MET parameters like Wind, Temperature, Visibility, Pressure, Cloud base height etc. on aircraft operations is very high. So the availability of accurate and reliable MET observations at aerodromes to support aircraft operations are of critical importance.

1.4 It is very much essential that all aviation MET instruments commissioned at public aerodromes for provisioning MET forecasts and warning(s) provide accurate data and information, free from errors to aeronautical users. The accuracy, integrity and reliability of the instruments are to be ensured at all times.

1.5 In order to maintain the accuracy and reliability of instrument(s) utilized for meteorological services in air navigation, following requirements for siting, calibration and maintenance of aviation MET instruments should be followed.

2. Applicability

This Advisory Circular (AC) is issued to provide guidance to meteorological service provider (IMD) on installation, calibration and maintenance requirements of meteorological equipment utilized for providing aviation meteorological services at public licensed aerodromes.

3. Siting /Installation requirements of MET equipment in aerodromes

IMD in co-ordination with aerodrome operator, should ensure that the MET equipment are suitably sited and installed at correct locations as outlined in following table.

Meteorological element observed	MET equipment	Representative conditions and Siting requirements
Surface wind speed and direction	Anemometer and wind vane/Ultrasonic Wind sensor	Representative conditions along the runway and touchdown zone The site must not be affected by buildings, etc., or by aircraft operations (e.g. jet efflux during taxiing). Siting as mentioned in DGCA CAR Section 9 Series M Part I Appendix 3 Para 4.1.1 and Appendix 2 of ICAO DOC 8896. *For detailed siting requirements refer to WMO Guide to Meteorological Instruments and Methods of Observation No. 8, Guide to Meteorological Observing and Information Distribution Systems for Aviation Weather Services WMO-No. 731.*
Air temperature	Thermometer	Representative condition of the aerodrome. *For detailed siting requirements refer to WMO Guide to Meteorological Instruments and Methods of Observation No. 8, Guide to Meteorological Observing and Information Distribution Systems for Aviation Weather Services WMO-No. 731*
Atmospheric pressure	Barometer	Representative condition of the aerodrome. *For detailed siting requirements refer to WMO Guide to Meteorological Instruments and Methods of Observation No. 8, Guide to Meteorological Observing and Information Distribution Systems for Aviation Weather Services WMO-No. 731*
Humidity / Dew point	Hydrometer	Representative condition of the aerodrome. *For detailed siting requirements refer to WMO Guide to Meteorological Instruments and Methods of Observation No. 8, Guide to Meteorological Observing and Information Distribution Systems for Aviation Weather Services WMO-No. 731*

Runway visual range(RVR)	Transmissometer and/or forward-scatter meter	Representative conditions of the runway. RVR should be assessed at a height of approximately 2.5 m (7.5 ft) above the runway for instrumented systems or assessed at a height of approximately 5 m (15 ft) above the runway by a human observer. RVR should be assessed at a lateral distance from the runway Centre line of not more than 120m. The site for observations to be representative of the touchdown zone should be located about 300 m along the runway from the threshold. The sites for observations to be representative of the mid-point and stop-end of the runway should be located at a distance of 1000 to 1500 m along the runway from the threshold and at a distance of about 300 m from the other end of the runway. The exact position of these sites and, if necessary, additional sites should be decided after considering aeronautical, meteorological and climatological factors such as long runways, swamps and other fog-prone areas. *For detailed siting requirements refer to DGCA CAR Sectin 9 Series M Part 1,, ICAO Manual of Runway Visual Range Observing and Reporting Practices Doc 9328.*
Height of Cloud base	Ceilometer	Representative conditions of the runway threshold. When instrumented systems are used for the measurement of the cloud amount and the height of cloud base, representative observations should be obtained by the use of sensors appropriately sited. For local routine and special reports, in the case of aerodromes with precision approach runways, sensors for cloud amount and height of cloud base should be sited to give the best practicable indications of the cloud amount and height of cloud base at the threshold of the runway in use. For that purpose, a sensor should be installed at a distance of less than 1200m (4000 ft) before the landing threshold. *For detailed siting requirements refer to Guide to Meteorological Observing and Information Distribution Systems for Aviation Weather Services WMO-No. 731*

4. Access to MET instruments within operational area of an aerodrome

The aerodrome operator should ensure that the MET personnel are able to access the MET sites within the aerodrome without any obstructions. The access road to the MET sites are to be kept clear of vegetation and other material for unobstructed measurement and maintenance of MET equipment.

The MET-officer in Charge of the aerodrome should co-ordinate with aerodrome operator for the access of all the MET sites for measurements and maintenance.

IMD while installing remote MET equipment within the operational area of airports should ensure redundancy in power supply in co-ordination with aerodrome operator for provisioning reliable aviation MET information.

5. Calibration requirement of aviation MET equipment:

Aviation MET Instruments should be calibrated as per periodicity given in table below:

Aviation MET equipment	Schedule of calibration
Anemometer and wind vane/Wind equipment	Once in six months
Thermometer	Once in a year
Barometer	Once in a year
Hygrometer	Once in a Year
Psychrometer	Once in a Year
RVR Instruments	Once in six months and prior to onset of FOG season
Ceilometers	Once in a Year

The calibration records of all MET instruments installed for aviation MET observations should be maintained at the aeronautical MET station at an airport.

The MET-Officer-in charge of the MET station at an airport is responsible for the timely calibrations of all MET instruments utilized for provisioning of Aviation MET services.

6. Maintenance requirements for aviation MET equipment

Inspection of field site must be carried out on fortnightly basis and Instrument inspection register is to be maintained at the MET office at the airport for recording deficiency, if any, and corrective action taken thereof.

This register should contain Fortnightly preventive maintenance and Quarterly controlled corrective maintenance records. The schedules and procedures should be made as per the SOPs/Circulars/guidelines issued by IMD. All aviation MET records should be maintained as per IMD record retention schedules issued from time to time.

The MET-Officer-in charge of an airport is responsible for the timely maintenance of all MET instruments utilized for provisioning of Aviation MET services and for upkeep of all records i.r.o MET instruments utilized for provisioning of Aviation MET services.

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